



ANALYSTLAB AFRICA
Experience Lab Internship Programme
Official Weekly Assignment
Week 2: FinTrust Digital Bank — Analyse & Prepare

Document Control

Week	Week 2
Project	FinTrust Financial Intelligence & Digital Banking Support Solution
Version	1.0
Issue Date	
Submission Deadline	End of Week 2 (Sunday, 11:59 PM WAT)

1. Week 2 Overview

Welcome to **Week 2 of the FinTrust Digital Bank Experience Lab Project.**

Week 1 focused on understanding the FinTrust business problem, reviewing the available project resources, defining your professional role, identifying objectives and success criteria, and creating an initial plan for the remaining project weeks.

Week 2 is where you move from planning into practical execution.

You are now expected to begin working directly with the FinTrust datasets, knowledge resources and technical tools relevant to your assigned track.

During this week, you will:

- analyse and prepare project data;
- develop practical analytical or technical outputs;
- apply professional tools;
- perform initial testing and evaluation;
- identify meaningful findings;
- document important decisions and limitations; and
- prepare your work for the next stage of the FinTrust project.

The FinTrust project is designed as a multidisciplinary data and AI project involving Data Analytics, Data Science, Machine Learning Engineering, Generative AI and Project Management. However, **Week 2 will be completed independently within your assigned track.**

Important Week 2 Policy

There is no formal cross-track collaboration requirement in Week 2.

You do not need to contact another intern or another track in order to complete this assignment.

Your priority this week is to demonstrate that you can independently apply the knowledge and plan developed in Week 1 to produce a substantial, practical output.

Formal integration and cross-track activities can be introduced in later stages of the project.

2. Project Resources

You must continue using the approved FinTrust resources provided by AnalystLab Africa.

Data Resources

- `FinTrust_Customer_Data.csv`
- `FinTrust_Transaction_Data.csv`
- `FinTrust_Data_Dictionary.xlsx`

GenAI Resource

- `FinTrust_Financial_Knowledge_Base.docx`

Project Resources

- `FinTrust Project Brief`
- `FinTrust 4-Week Master Roadmap`

Data Disclaimer

All FinTrust customers, transactions, labels and financial information are **synthetic and created for educational purposes only.**

The `Risk_Review_Flag` must not be presented as a real fraud determination or real banking risk decision.

3. General Week 2 Requirements

Regardless of your track, you must complete the following.

A. Apply Your Week 1 Plan

Review the plan you created in Week 1.

Identify:

1. What did you originally plan to accomplish?
2. Which activities are you now implementing?
3. Have any aspects of your approach changed?
4. Why did you make any changes?

You should not simply repeat your Week 1 plan. You should demonstrate **progress from planning into execution**.

B. Use the Approved Project Resources

Use the FinTrust resources relevant to your assigned track.

You must be able to explain:

- which resources you used;
- how you used them;
- what information they provided;
- any data or resource limitations you encountered.

C. Use the Required Tools

You must use the tools specified for your assigned track.

The objective is not to use a tool simply for the sake of using it.

You should demonstrate that you can use the appropriate tool to perform a meaningful project activity.

D. Produce Practical Outputs

Your Week 2 submission must contain actual work.

Depending on your track, this may include:

- cleaned datasets;
- SQL queries;
- Python analysis;
- dashboards;
- machine-learning models;
- feature engineering;
- ML pipelines;
- software code;
- AI assistant prototypes;
- evaluation results;

- project management documents.

E. Test or Evaluate Your Work

Do not submit an output without checking its quality.

Document:

Work Completed → Test/Evaluation → Finding → Action/Improvement → Result

The level of testing should be appropriate to your assigned track.

F. Document Important Decisions

For important decisions, explain:

- what you decided;
- why you made the decision;
- what evidence supported the decision;
- how the decision affected your work.

G. Identify Limitations

Document relevant:

- assumptions;
- data-quality issues;
- technical limitations;
- methodological limitations;
- risks;
- unresolved issues.

4. DATA ANALYTICS TRACK

Task: FinTrust Customer & Transaction Data Analysis

Your role is to transform FinTrust's customer and transaction data into meaningful business intelligence that can support management decision-making.

Week 1 focused on business understanding, data profiling, analytical questions, KPI planning and dashboard planning. Week 2 requires you to begin **actually performing the analysis**.

Part A — Data Preparation & Quality Assessment

Using:

- `FinTrust_Customer_Data.csv`
- `FinTrust_Transaction_Data.csv`

- FinTrust_Data_Dictionary.xlsx

use **Microsoft Excel** to perform an initial data-quality assessment.

Document:

1. Number of records.
2. Number of columns.
3. Missing values.
4. Duplicate records.
5. Incorrect or inconsistent values.
6. Data-type issues.
7. Unusual values.
8. Potential outliers.
9. Relationship between the Customer and Transaction datasets.
10. Any issues that could affect analysis.

Where appropriate, create cleaned versions of the datasets.

Required Tool

Microsoft Excel

Deliverable

Data Quality & Cleaning Workbook

Part B — SQL Business Analysis

Load the appropriate data into a SQL environment.

Use SQL to answer at least **8 meaningful business questions**.

Your questions should cover areas such as:

- customer behaviour;
- transaction activity;
- transaction value;
- transaction type;
- transaction channel;
- transaction status;
- customer segments;
- risk-review patterns.

For each question provide:

Business Question SQL Query Result Business Interpretation

Do not submit SQL queries without explaining what the results mean.

Recommended SQL Tools

You may use an appropriate SQL environment such as:

- MySQL
- PostgreSQL
- SQL Server
- SQLite
- DuckDB

Deliverable

FinTrust_Week2_SQL_Analysis.sql

Part C — Python Exploratory Data Analysis

Use Python to perform exploratory analysis.

Your analysis should investigate areas such as:

- customer segments;
- transaction types;
- transaction amounts;
- transaction channels;
- transaction status;
- international transactions;
- customer behaviour;
- risk-review patterns.

Create at least **5 meaningful visualizations**.

For each important visualization, explain:

What does the visual show?

Why is it important?

What business insight can be derived from it?

Required Tools

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook or Google Colab

Deliverable

FinTrust_Week2_Data_Analysis.ipynb

Part D — Power BI Dashboard Development

Use Power BI to create an **initial management dashboard**.

Your dashboard should contain appropriate KPIs.

Examples include:

- Total Customers
- Total Transactions
- Total Transaction Value
- Average Transaction Value
- Transaction Success Rate
- Risk Review Rate

Include appropriate visualizations for:

- customer segments;
- transaction types;
- transaction channels;
- transaction trends;
- transaction status;
- risk-review patterns.

Include appropriate filters or slicers.

Required Tool

Microsoft Power BI

Deliverable

FinTrust_Week2_Analytics_Dashboard.pbix

Part E — Business Insights

Identify at least **5 meaningful business findings** from your analysis.

Use the structure:

Finding → Evidence → Business Meaning

Avoid simply stating:

"The chart shows that X is higher."

Instead explain what the finding could mean for FinTrust's business understanding or decision-making.

Recommended Tools

- Microsoft Excel
- SQL
- Python/Pandas
- NumPy
- Matplotlib
- Seaborn
- Power BI
- Jupyter Notebook/Google Colab

Data Analytics Deliverables

1. Data Quality & Cleaning Workbook
2. SQL Analysis
3. Python EDA Notebook
4. Power BI Dashboard
5. Business Findings
6. Week 2 Documentation

5. DATA SCIENCE TRACK

Task: FinTrust Risk Review Predictive Model — Initial Development

Your role is to use machine learning to investigate whether transaction information can support prediction of the synthetic `Risk_Review_Flag`.

Week 1 focused on defining the predictive problem, assessing the target, identifying candidate features, developing hypotheses and creating an initial modelling plan.

Week 2 moves into **data preparation, exploratory analysis, feature engineering and baseline model development**.

Part A — Data Preparation

Using Python:

1. Load the customer dataset.
2. Load the transaction dataset.
3. Join the datasets using `Customer_ID`.
4. Inspect missing values.
5. Check duplicates.
6. Review data types.
7. Investigate unusual values.
8. Examine the target distribution.

9. Identify variables that may be inappropriate for modelling.
10. Document your data preparation decisions.

Part B — Exploratory Data Analysis

Perform EDA on the modelling dataset.

Investigate relationships between the target and relevant variables.

Your analysis should consider areas such as:

- transaction amount;
- transaction type;
- channel;
- device type;
- location;
- international transactions;
- transaction status;
- customer segment;
- account type;
- digital engagement.

Create at least **8 meaningful visualizations**.

Part C — Feature Engineering

Create at least **8 useful predictive features**.

Possible examples include:

- transaction amount bands;
- transaction hour;
- transaction day;
- customer tenure;
- digital engagement category;
- international transaction indicator;
- customer transaction count;
- transaction frequency.

For each feature, document:

Feature How It Was Created Why It May Matter

Do not create features without explaining their relevance.

Part D — Baseline Model Development

Develop at least **one baseline classification model**.

Appropriate algorithms may include:

- Logistic Regression;
- Decision Tree;
- Random Forest;
- another justified classification algorithm.

You must:

1. Prepare the modelling data.
2. Split the data appropriately.
3. Train the model.
4. Generate predictions.
5. Evaluate the model.

Part E — Model Evaluation

Use appropriate classification metrics.

Your evaluation should consider:

- Accuracy;
- Precision;
- Recall;
- F1-score;
- Confusion Matrix;
- ROC-AUC where appropriate.

Explain why the selected metrics are appropriate for your problem.

Part F — Initial Model Interpretation

Identify important variables or features influencing the model.

Document:

Feature → Evidence → Interpretation

Also document:

- model limitations;
- data limitations;
- possible sources of bias;
- areas requiring further investigation.

Important

The `Risk_Review_Flag` is a **synthetic educational target**.

It must not be described as evidence of real fraud or used to make real financial decisions.

Recommended Tools

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook
- Google Colab
- Git/GitHub

Data Science Deliverables

1. Prepared Modelling Dataset
2. EDA Notebook
3. Feature Engineering Documentation
4. Baseline Model
5. Model Evaluation
6. Model Interpretation
7. Limitations
8. Week 2 Project Documentation

6. MACHINE LEARNING ENGINEERING TRACK

Task: FinTrust ML Workflow — Initial Implementation

Your role is to begin transforming the planned machine-learning architecture from Week 1 into a **reproducible technical workflow**.

Week 1 focused on architecture, technical requirements, repository structure, risks and testing strategy.

Week 2 focuses on **initial implementation**.

Part A — Project Structure

Create a professional project repository.

Explain the purpose of the folders and files you create.

Part B — Data Validation

Develop an initial data-validation process.

Test for issues such as:

- missing values;
- incorrect data types;
- unexpected categories;
- empty datasets;
- invalid inputs;
- unexpected columns.

Document what happens when invalid data is detected.

Part C — Preprocessing Workflow

Implement an initial preprocessing workflow.

Where appropriate, include:

- missing-value handling;
- categorical encoding;
- numerical processing;
- feature preparation.

The workflow should be reproducible.

Part D — Model/Prediction Workflow

Develop an initial workflow following:

Data → Validation → Preprocessing → Feature Preparation → Model → Prediction → Output

Your implementation should demonstrate how data moves through the workflow.

Part E — Technical Testing

Create at least 5 **technical tests**.

Examples:

1. Valid input.
2. Missing values.
3. Unexpected category.
4. Incorrect data type.
5. Empty input.

Document:

Test → Expected Result → Actual Result → Status

Part F — Reproducibility

Ensure that another person can understand how to run your project.

Include:

- `README.md`;
- `requirements.txt`;
- installation instructions;
- execution instructions;
- project structure;
- assumptions;
- known limitations.

Part G — Git/GitHub

Use Git/GitHub to manage your project.

Demonstrate:

- meaningful commits;
- organised repository structure;
- updated README;
- appropriate file organisation.

Optional Advanced Component

Where appropriate, develop or document an initial **API/service concept** for model prediction.

Recommended Tools

- Python
- VS Code
- Jupyter Notebook
- Git
- GitHub
- Scikit-learn
- API concepts
- Testing frameworks/concepts

ML Engineering Deliverables

1. Project Repository

2. Data Validation Component
3. Preprocessing Workflow
4. Prediction Workflow
5. Technical Tests
6. requirements.txt
7. README
8. Git/GitHub Evidence
9. Technical Documentation

7. GENERATIVE AI TRACK

Task: FinTrust Customer Support Assistant — Initial Development & Evaluation

Your role is to develop an initial customer-support assistant using the approved **FinTrust Financial Knowledge Base**.

Week 1 focused on knowledge-base assessment, assistant scope, initial prompts, evaluation questions and proposed architecture.

Week 2 moves into **initial assistant development and evaluation**.

Part A — Knowledge Base Implementation

Use:

`FinTrust_Financial_Knowledge_Base.docx`

Your assistant must use the approved information as its knowledge source.

The Knowledge Base includes information relating to:

- account access;
- transaction issues;
- transfers;
- cards and payments;
- fees and limits;
- escalation;
- safe responses.

Part B — Prompt Development

Develop at least **8 prompts** covering different customer-support scenarios.

Your prompts should include scenarios such as:

- account-access support;
- failed transactions;
- reversed transactions;
- transfer issues;
- unrecognised transactions;
- card/payment issues;
- fees or limits;
- unsupported questions.

Part C — Initial Assistant Development

Develop a working prototype following the general flow:

User Question → Prompt → Knowledge Retrieval/Context → LLM → Response

Where a retrieval system is used, clearly document how the relevant knowledge is retrieved.

Part D — Grounding Tests

Test whether the assistant stays within the approved FinTrust information.

Include questions that:

- can be answered from the Knowledge Base;
- cannot be answered from the Knowledge Base;
- require escalation;
- may tempt the assistant to invent information.

Part E — Safety Tests

Test scenarios involving:

- passwords;
- PINs;
- one-time passcodes;
- authentication codes;
- account-specific information;
- disputed transactions;
- security concerns;
- personalised financial advice.

The assistant must not request or expose sensitive credentials and must not provide personalised financial, investment, lending or legal advice.

Part F — Evaluation

Create at least **15 evaluation questions**.

For each test, record:

Question	Expected Behaviour	Actual Response	Result	Issue	Action
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Part G — Initial Refinement

Based on your evaluation:

- identify weak responses;
- identify grounding issues;
- identify unsafe responses;
- improve prompts or retrieval;
- re-test the affected scenarios.

Critical Rule

The assistant must **not invent FinTrust policies, fees, limits, timelines, contact details or procedures** that are not contained in the approved Knowledge Base.

When information is unavailable, the assistant should acknowledge the limitation and follow the defined escalation approach.

Recommended Tools

- Python
- Jupyter Notebook/Google Colab
- VS Code
- LLM/API
- Git
- GitHub
- Appropriate GenAI frameworks where applicable
- Retrieval/embedding tools where applicable

Generative AI Deliverables

1. Knowledge Base Implementation
2. Prompt Library
3. Initial Assistant Prototype

4. Evaluation Dataset
5. Grounding Tests
6. Safety Tests
7. Evaluation Results
8. Refinements
9. Documentation
10. GitHub Repository where applicable

8. PROJECT MANAGEMENT TRACK

Task: FinTrust Project Execution & Progress Management

Your responsibility is to manage the FinTrust project during the transition from planning into practical execution.

Week 1 focused on the project charter, WBS, four-week timeline, initial dependencies, risk register and reporting plan.

Week 2 focuses on **monitoring actual project execution**.

Part A — Week 2 Progress Tracking

Update the Week 1 plan based on actual progress.

Track:

- planned activity;
- actual activity;
- status;
- progress;
- expected completion;
- issues.

Part B — Updated Timeline

Update the four-week project timeline based on Week 2 progress.

Document any changes and explain why they were made.

Part C — Risk Management

Review the Week 1 risk register.

Identify:

- risks that remain active;
- risks that have changed;
- risks that have been resolved;
- new risks discovered during Week 2.

Part D — Issue Log

Create an issue log containing:

Issue Impact Priority Action Owner Status

Part E — Dependency Monitoring

Continue monitoring dependencies identified in Week 1.

Important: identifying or monitoring a dependency does not mean that formal collaboration is required.

Part F — Decision Log

Record important project decisions.

Decision Reason Impact Date

Part G — Week 2 Status Report

Prepare a concise project status report covering:

1. Overall project progress.
2. Activities completed.
3. Activities in progress.
4. Delayed activities.
5. Major risks.
6. Major issues.
7. Decisions made.

8. Project health.
9. Recommended Week 3 priorities.

Recommended Tools

- Microsoft Excel
- Google Sheets
- Microsoft Word
- Google Docs
- PowerPoint
- Google Slides
- Trello/Notion/Jira where appropriate

Project Management Deliverables

1. Updated Progress Tracker
2. Updated Timeline
3. Risk Register
4. Issue Log
5. Dependency Register
6. Decision Log
7. Week 2 Status Report

9. WEEK 2 TESTING & QUALITY REQUIREMENT

Every intern must demonstrate that they have reviewed the quality of their work.

Use the following structure where applicable:

Stage	Description
Work Completed	What did you produce?
Test/Evaluation	How did you check it?
Finding	What did you discover?
Action/Improvement	What did you change or decide?
Result	What happened after the change?

The level and type of testing should be appropriate to your track.

10. ASSUMPTIONS, LIMITATIONS, RISKS & DECISIONS

Review your Week 1 assumptions, limitations, risks and dependencies.

Document:

- which remain relevant;
- which have been resolved;
- which have changed;
- new issues discovered during Week 2;
- their potential impact;
- how they should be addressed.

Do not create a separate report if this information can be incorporated into your main track-specific submission.

11. WEEK 2 PROJECT SUMMARY

Prepare a concise **Week 2 Project Summary** covering:

1. What you planned to accomplish in Week 2.
2. What you actually completed.
3. Tools used.
4. Key findings or development outcomes.
5. Major challenges encountered.
6. Important decisions made and why.
7. Any changes to your Week 1 approach.
8. Testing or evaluation performed.
9. Key limitations.
10. Remaining work.
11. Your proposed focus for Week 3.

The summary should complement your main project documentation rather than simply repeat it.

12. WEEK 2 SUBMISSION

Each intern submits an **individual Week 2 submission**.

Your submission must include:

Required

- Track-specific Week 2 output.
- Week 2 Project Summary.

- Relevant analysis/code/notebooks.
- Supporting datasets or processed datasets where applicable.
- Testing/evaluation evidence.
- Documentation.
- Relevant screenshots or evidence.
- GitHub repository where applicable.

Google Drive

Upload the required Week 2 files to your designated project folder.

Ensure:

- correct files are uploaded;
- files are properly named;
- folders are organised;
- mentor access is available.

GitHub

Where applicable:

- update your repository;
- commit Week 2 work;
- update `README.md`;
- maintain an organised project structure;
- ensure links/files are accessible.

13. PROFESSIONAL DEVELOPMENT REQUIREMENT

Continue documenting your FinTrust project journey professionally.

Interns should:

- maintain an organised project folder/repository;
- document important decisions;
- maintain appropriate version control;
- update their README;
- present their work professionally;
- share meaningful project progress on LinkedIn and X where appropriate;
- tag **AnalystLab Africa** on LinkedIn;
- tag **@analystlabafrik** on X where appropriate;
- use **#AnalystLabAfrica** where appropriate.

Any publicly shared work must accurately represent the intern's actual work and progress.

14. ASSESSMENT CRITERIA

Criteria	Weight
Application of Week 1 Foundation	10%
Track-Specific Analysis & Development	30%
Technical/Analytical Quality	25%
Tool Application & Practical Implementation	15%
Problem-Solving & Decision-Making	10%
Documentation & Organization	10%
Total	100%

Assessment Focus

Your Week 2 work will be assessed based on your ability to:

- apply your Week 1 foundation;
- perform meaningful practical work;
- use the required tools appropriately;
- produce technically or analytically sound outputs;
- demonstrate problem-solving;
- test or evaluate your work;
- document your decisions and limitations;
- present your work professionally.

Formal cross-track collaboration is not assessed in Week 2.

Professional Development Requirement

As part of building your professional portfolio:

- Update your GitHub repository with your Week 1 project.
- Publish a LinkedIn post describing your project
- Publish an X (Twitter) post summarizing your project.
- Tag **AnalystLab Africa** on LinkedIn.
- Tag **@analystlabafri**c on X.
- Include the hashtag **#AnalystLabAfrica** on both platforms.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.